2025 Narrative Technical Progress for Improving Generalization of Neural Networks for Physics Domains

Technical Objective

TLDR: Our objective is to demonstrate the superior accuracy, generalization, and computational efficiency of novel deep learning (DL) architectures for solving complex, nonlinear physics-based problems critical to Naval operations. To do this, we will develop and validate neural surrogate models and neural operators integrated with physics-based feature engineering, specifically for underwater acoustics, electromagnetic propagation, and structural health monitoring applications. This will provide the Navy with real-time, highly accurate predictive capabilities for critical maritime physics applications, accelerating decision-making, reducing operational costs, and enhancing tactical advantages in contested environments.

Deep learning (DL) techniques have demonstrated substantial success in computer vision and natural language processing tasks. However, critical gaps persist in applying these methods to physics-based problems essential to Naval operations, including underwater acoustics, electromagnetic (EM) propagation, and structural health monitoring (SHM). This research addresses these limitations by developing novel DL architectures and training methodologies designed to achieve enhanced accuracy across the complex, nonlinear problem spaces characteristic of maritime environments. The proposed work directly supports the Navy's Distributed Maritime Operations (DMO) concept and Marine Corps Expeditionary Advanced Base Operations (EABO), which require resilient, real-time decision-making capabilities in contested environments.

This research directly supports Navy operational priorities by focusing on three specific Navy-relevant physics applications involving Helmholtz wave equation solutions: underwater acoustics, EM propagation, and SHM.

1. Current submarine sonar systems rely on computationally intensive Naval Standard Parabolic Equation (NSPE) simulations to predict acoustic performance and identify vulnerabilities. Operators require accurate predictions across varied frequencies and geometries to counter adversarial sonar countermeasures. This research will develop neural operators incorporating governing equations (e.g., wave propagation, boundary layer interactions), ensuring physical consistency. Hybrid models combining DL with reduced-order physics solvers are designed to accelerate predictions by two to three orders of magnitude compared to NSPE, enabling real-time sonar performance assessments during missions. This capability directly supports the Navy's Anti-Submarine Warfare (ASW) priorities by potentially improving detection probabilities against quiet diesel-electric submarines in littoral zones.
2. Shipboard Meteorology and Oceanography (METOC) personnel rely on Tactical Decision Aids (TDAs) to assess EM radio frequency (RF) propagation characteristics. Current numerical weather prediction (NWP) systems require multiple hours to generate forecasts, limiting responsiveness to rapidly evolving environmental conditions. This work will develop neural surrogate models trained on Navy Coupled Ocean/Atmosphere Mesoscale Prediction System (COAMPS) data, targeting two orders of magnitude speedup while maintaining prediction errors below 5% in refractivity calculations.
3. Scheduled inspections of DoD platforms incur substantial annual costs and reduce platform availability for operational missions. This research will develop SHM neural networks designed to generalize from controlled laboratory experiments to battlefield conditions, including shock loads, wide temperature ranges, and corrosion effects. By integrating neural operators with physics-based feature engineering, the models will detect microcracks and delamination under variable operational loads, enabling predictive maintenance on complex equipment in real-world environments. This directly supports the Marine Corps' Force Design 2030 goal of distributed sustainment for small-unit operations.

The research is expected to yield substantial reductions in computational resource requirements, with projected acceleration factors of two to three orders of magnitude in physics simulations compared to traditional numerical methods. Additionally, enhanced generalization capabilities across diverse physical environments should substantially improve system accuracy in complex and varied operational scenarios. Enhanced sonar prediction accuracy has the potential to improve ASW effectiveness, while real-time EM propagation models may provide capabilities to counter adversary electronic warfare (EW) tactics. While focused on Navy systems, this work has potential applications benefiting other DOD services and civilian maritime industries.

Our research addresses critical gaps in applying deep learning methodologies to physics problems relevant to naval operations. By developing techniques designed to achieve enhanced accuracy and generalization across diverse physical environments, this work will enable more efficient design processes, improved operational capabilities, and reduced operational costs across multiple naval platforms. The enhanced predictive capabilities for underwater acoustics and electromagnetic signatures are expected to provide tactical advantages in contested environments.

Technical Approach

Executive Summary: This research program addresses three fundamental challenges in physics-informed machine learning: (1) inadequate generalization of standard deep learning techniques for physics applications, (2) requirements for models that generalize across diverse physical environments and governing equations, and (3) computational limitations of traditional numerical physics solvers. Our integrated approach combines physics-guided neural architectures, domain-specific generalization strategies, and scalable neural operator frameworks to achieve robust, high-fidelity predictions across complex physical systems. The research focuses primarily on wave propagation phenomena governed by the Helmholtz equation, with applications spanning sonar acoustic modeling, radio frequency (RF) propagation, and structural health monitoring. Neural operator frameworks will be the primary technical focus throughout the rest of this project.

Problem Definition and Scientific Framework: The central technical challenge involves developing generalizable deep learning methodologies for solving partial differential equations (PDEs) across diverse physics domains, with emphasis on wave propagation phenomena. Traditional deep neural networks (DNNs) exhibit poor generalization when applied to physics problems, demonstrating degraded accuracy when encountering environmental conditions, frequencies, or boundary conditions absent from training distributions. Our FY24 investigations demonstrated that standard generalization techniques from computer vision (e.g., mixup [1], Sharpness-Aware Minimization [2]) prove ineffective for physics applications, necessitating domain-specific methodologies incorporating fundamental physical principles.

Research Hypothesis: Neural operator frameworks incorporating physics-informed constraints and domain-specific generalization strategies will demonstrate superior computational efficiency and accuracy relative to traditional numerical solvers and conventional deep learning approaches across three critical naval applications: sonar acoustic modeling, RF propagation, and structural health monitoring.

Application Background: In the past few years, the research community has witnessed exponential growth in neural network integration with physics for solving PDEs. This project builds upon existing efforts to implement neural network surrogates for three task domains of strategic importance to NRL researchers, each utilizing wave propagation phenomena governed by the Helmholtz equation: sonar acoustic modeling, RF propagation, and structural health monitoring. The confidential nature of the data and direct relevance to naval operations necessitate conducting this research exclusively at NRL facilities.

Acoustics: The overall objective of the Physics-based Acoustic machine Learning for In-Situ Applications (PALIS)effort is to develop modern, physics-informed deep neural networks (DNNs) and generative networks to advance acoustic simulation technologies available to Naval Anti-Submarine Warfare (ASW) decision-makers. Our goal is to enable operators, both land and sea-based, to better exploit critical environmental acoustic knowledge. To achieve this, we have developed, implemented, and tested physics-based DNNs to solve computationally expensive acoustic problems quickly and accurately. We have also used advanced generative methods to reduce the digital footprints of simulated acoustic and ocean fields. Finally, we have developed and applied acoustic metrics to evaluate the technologies.

Our results represent an important advance in the fields of physics-informed DNNs and acoustic simulation modeling. Our resulting algorithms will transition to tactical decision aids (TDAs) in place of the current expensive algorithms. Such a capability will significantly improve the combat effectiveness of US submarines (and others) by providing faster, more complete predictions with uncertainty in operational time frames.

The theoretical work involved recasting propagation physics into physics-informed neural networks and developing generative methods that maintain acoustic integrity. The experimental work involved extensive simulation experiments to identify sensitivity, train the systems, and test resulting methods against their current counterparts.

The performers of the 6.2 PALIS project developed two neural network-based methods for estimating transmission loss (TL) from ocean properties for limited ranges of ocean conditions and source depths/frequencies. Their implementations were the baseline for the acoustics domain. In this project, we have focused on developing methods for training NNs such that they produce accurate 2D TL maps across time, and for a broad spectrum of ocean properties (i.e., temperature, salinity, sound speed), sound source depths, receiver locations, and sound source frequencies.

In summary, our work in the acoustics domain has resulted in the development of modern, physics-informed DNNs and generative networks that will significantly improve the combat effectiveness of US submarines and other naval forces. Our algorithms have been tested and evaluated against current methods, and we have made significant progress in developing methods for training NNs to produce accurate 2D TL maps for a broad range of ocean conditions and source characteristics.

Meteorology: The Navy frequently operates within the marine atmospheric surface layer (MASL), a dynamic environment characterized by varying vertical profiles of thermodynamic variables such as temperature, humidity, pressure, and wind. These gradients in thermodynamic variables cause changes to the refractive environment, most notably ducts, which are atmospheric refractive phenomena caused by vertical gradients of humidity and temperature. Ducts act as trapping layers, leading to anomalous propagation such as extended signal range within the duct and signal loss above the duct for radio frequency (RF) technologies. Accurately modeling the refractive environment's impact on RF technologies is essential for national defense.

To evaluate RF propagation in the MASL, we use physics-based simulations to calculate the propagation factor across the entire atmospheric domain. Modified refractivity is routinely used to describe the refractive environment, as it accounts for the curvature of the Earth. Changes in the gradient of the modified refractivity profile are indicative of how RF energy will either follow or depart from Earth's surface. The propagation factor is a crucial metric used to describe environmental attenuation, which can be calculated as the magnitude of the propagation factor or through the calculation of propagation loss.

Estimating radio frequency (RF) propagation in the atmosphere has many similarities to estimating transmission loss (TL) in the underwater acoustics domain. Therefore, we anticipated that the neural network (NN) solutions developed in PALIS can be used with the Navy's Coupled Ocean/Atmosphere Mesoscale Prediction System (COAMPS®) NWP data. The acoustics software was modified for this data and tested in the new domain. This modified implementation was adapted for our meteorological baseline. Then, we investigated modifying these algorithms to produce accurate RF propagation maps across location, time, and a spectrum of atmospheric properties (i.e., humidity, temperature, pressure), trapping conditions (sub-refractions, standard-refractions, super-refraction, and trapping), and relevant RF source heights, receiver locations, and RF source frequencies.

Structural health monitoring: Our Material-Systems Intelligence program was formed on our notion that ML and AI have had substantial success in the commercial sector (machine vision for self-driving cars, generative AI, etc.) but have not been well implemented in the world of materials science. Our program objective is to develop cutting-edge solutions by taking a physics-first approach to scientific machine learning with the goal of integrating experimental data, computational modeling, and ML for a use case of damage detection on Naval-relevant structural materials.

We have identified two major gaps in this research field. The first major gap we saw in the field was that most other researchers were training on just computationally generated data which is too pristine and doesn't work for complex systems. Our approach was to generate very large experimental datasets and couple those with computational modeling in specific ways to train neural networks.

Of course, it is not feasible to have thousands of metal plates with various degrees of damage, so we came up with a method of using "Simulated" damage in the form of contact loads. Because experimental datasets in the tens of thousands are likely needed for increased prognostic capabilities (for example, moving from detection, to localization, to characterization, to estimation of remaining useful life), an automated robotic gantry was designed and assembled to move the contact loads all over the plates. This is all run with a custom Matlab code that runs the robotic movements, acquires the lamb wave signals, and saves the data with a nomenclature that is pulled in by our ML algorithms as labeled training data.

The second gap we saw was on the ML side. Researchers in the damage detection field are taking the time series data and transforming that into a form that can be used on established ML/ANN algorithms that are excellent at image analysis – such as those for computer vision and self-driving cars. They are ignoring the physics of the system. What we set out to do was to write the Neural Network algorithms to specifically leverage the physics of the problem and incorporate the time-frequency sensor data directly. So far, our colleagues doing the algorithm development have made tremendous progress.

The time-series data, which is composed of wavepackets, lends itself to a sparse Gabor representation; by providing the network with a mixed time-frequency domain input, we can leverage stacked convolutional layers to build a global representation from locally extracted features while both discarding noise and mitigating the tendency of neural networks to wash out relevant high-frequency information. With the implementation of this, we quickly achieved 100% detection accuracy.

This task domain differs from the prior two domains. The performers for the 6.1 MINC project (i.e., Smith and Rudolf) are developing a real-time NN system for monitoring material damage on naval platforms. We propose developing NN training methods that enable the developed systems to perform with high accuracy over a large variety of metallic surfaces (e.g., steel, aluminum, composites), across all configurations of sensor locations, for complex structural geometries, interacting boundaries, non-linear wave dynamics (dispersion, interference, attenuation), and a near infinite population of possible damage characteristics.

In summary, our work in the structural health monitoring domain has focused on developing cutting-edge solutions by taking a physics-first approach to scientific machine learning. We have identified two major gaps in this research field and have proposed a method for generating very large experimental datasets using simulated damage and an automated robotic gantry. We have also proposed a new approach to writing Neural Network algorithms that specifically leverage the physics of the problem and incorporate the time-frequency sensor data directly. Our progress had been substantial, and we have achieved 100% detection accuracy in our experiments. Our methods will enable the developed systems to perform with high accuracy over a large variety of metallic surfaces, across all configurations of sensor locations, for complex structural geometries, interacting boundaries, non-linear wave dynamics, and a near-infinite population of possible damage characteristics.

Expected Accomplishments:

- Establishment: Robust generalized DL models accurately predicting outcomes within unseen physical environments previously unattainable via conventional means!
- Contribution: Advancing scientific ML frameworks, developing scalable solutions universally applicable across diverse physic-domains unlocking unprecedented predictive potential, benefiting multiple military/civilian sectors equitably!

References

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Technical Progress

During FY25, we identified that recent developments in neural operators for physics-guided neural networks [1] represent a significant advancement in machine learning applications to scientific computing and align directly with this program's objectives. Neural operators constitute a generalization of conventional neural networks, specifically designed to learn mappings between infinite-dimensional **function spaces**. While traditional neural networks learn mappings between finite-dimensional spaces (transforming discrete numerical inputs to discrete numerical outputs), neural operators learn transformations from complete input functions to complete output functions. As detailed below, neural operators demonstrate potential for learning functional mappings across all conditions and parameters of specific partial differential equations (PDEs) or ordinary differential equations (ODEs), such as the Helmholtz wave equation, **potentially enabling universal physics solvers that require single training cycles yet function across diverse parameter spaces.**

This section presents our advancements in enhancing the generalization capabilities of neural networks for physics applications. Our research focused on developing and evaluating novel techniques that improve neural network applicability in complex physics problems, with particular emphasis on increasing predictive accuracy and robustness across physical conditions not explicitly represented in training datasets. Through systematic experiments and evaluations, we investigated the efficacy of our proposed methods in improving neural network performance for physics-related tasks.

This section is organized as follows: Section 1 describes general progress applicable across all three physics domains; Section 2 details advances in underwater acoustics; Section 3 presents our publication on modified refractivity prediction in the marine atmospheric surface layer (MASL); and Section 4 describes progress in structural health monitoring (SHM).

1. Progress that is applicable in all three domains

1.1 Neural Operators

Neural network-based surrogate models for wave physics simulation often exhibit limited generalization beyond training conditions. We investigated neural operators – a class of deep learning models that learn mappings between function spaces – to improve the generalization of neural surrogates for the Helmholtz equation governing wave propagation in inhomogeneous media (underwater acoustics, electromagnetic waves in atmospheric environments, and elastic waves in metallic structures). The cylindrical Helmholtz equation, a frequency-domain wave equation, is fundamental to modeling these applications. However, solving this equation for varying environmental parameters (e.g., sound-speed profiles, bathymetry in underwater acoustics) using traditional physics simulators such as NRL's NSPE requires substantial computational resources.

We conducted a comprehensive review of foundational operator learning methods, including Fourier-based approaches, graph-based methods, and multi-scale U-Net architectures, along with recent advances in neural operators applied to elastic wave equations. Our analysis focused on their capabilities for handling high-frequency wave components, complex boundary conditions (free surface, seabed interfaces, Sommerfeld radiation), and point-source singularities. Published literature demonstrates that neural operators provide improved generalization, accelerated inference, and enhanced scalability for wave propagation tasks.

Following a systematic comparison of foundational neural operators, including DeepONets and Fourier Neural Operators (FNOs) with their variants, we determined that the Neumann Series Neural Operator (NSNO) [2] approach for solving heterogeneous Helmholtz equations in underwater acoustic modeling provided the optimal foundation for our experiments. The NSNO architecture draws inspiration from the analytical Neumann series expansion of the Helmholtz solution operator. By reformulating the solution as a series that conceptually accounts for multiple scattering events in inhomogeneous media, the designers created a network that mimics this mathematical structure: a U-Net-based neural operator (termed "UNO") approximates each series term. The architecture incorporates multi-scale skip connections to capture the broad wavelength range present in high-frequency solutions.

In benchmark evaluations, NSNO demonstrated superior performance compared to standard FNO on 2D Helmholtz problems, achieving at least 60% error reduction and improved stability for large wavenumbers. Additionally, NSNO served effectively as a surrogate in inverse scattering applications (predicting source locations from observed signals), producing forward solutions with sufficient accuracy that inverse problem results were comparable to traditional solvers while requiring substantially reduced computational cost. This performance suggests neural operators can achieve accuracy levels necessary for sensitive forward or inversion tasks while providing significant computational efficiency gains.

The code implementation is currently in progress. Subsequent steps include conducting generalization experiments on the operator network, with initial application to the underwater acoustic domain, followed by extension to the SHM domain.

1.2 Contrastive Language Image Pre-training (CLIP) for creating a “foundation model” for a physics domain

We investigated training foundation-like models on comprehensive domain data, followed by fine-tuning for specific regions. This methodology resembles unsupervised CLIP training [3], which connects language and images for text-to-image generation models. CLIP trains models to understand text-image relationships by learning to predict corresponding image-text pairs. We hypothesized that pretraining our generative model using CLIP principles to predict SSP-TL pairs would create models with robust understanding of relationships between physical parameters and their visual representations, enabling application across diverse tasks and environments. Our implementation trained two encoders: input-to-representation (e.g., SSP to representation for PALIS) and output-to-representation (e.g., TL to representation for PALIS). During fine-tuning, encoder weights remained fixed while only the decoder underwent training.

We hypothesized that CLIP pretraining would improve generalization to new data by learning rich representations of underlying physical processes through training on large, diverse datasets. This capability is essential for physics domain problems where generalization to unseen data is critical for real-world applications. Additionally, we expected improved efficiency through more compact representations of physical processes, enabling accurate predictions with reduced training data and computational resources.

Another hypothesized benefit of pretraining a generative model using CLIP was that it would improve the model's efficiency. By training the model on a large and diverse dataset, it would learn a more compact representation of the underlying physical processes. This should enable the model to make accurate predictions with less training data and less computational resources, which can be especially important in the context of large-scale physics simulations. In summary, we expected that pretraining a generative model using CLIP had the potential to provide a number of benefits for solving physics domain problems, including improved generalization, efficiency, and versatility.

Extensive FY25 experiments demonstrated that expected benefits were not realized. Comparative analysis between pretrained models and locally trained models showed that pretrained model results were occasionally equivalent but generally inferior to local model training. This pattern persisted across a wide range of decreasing training dataset sizes where pretrained models were expected to demonstrate superiority. The pretrained model did not achieve the universal applicability we anticipated.

2. Underwater Acoustics Domain

This section discusses FY25 progress, which builds upon FY24 developments in the implementation approach, data preparation, performance measures, neural network architecture, loss functions, hyperparameter optimization, data augmentation, and intensity imbalance mitigation.

2.1 Utilize an approximate solution that is much faster to compute

In summary, our work had focused on utilizing an approximate or simplified solution that is much faster to compute and can be computed on the fly to improve the generated TLs in cases where the original PALIS code was not producing results of sufficient quality. Our experiments have illustrated that this method works very well, and we have modified the PALIS code to accept inputs with multiple channels. We found that the use of acoustic ray-traces as a second input channel performs as well or better than a low-resolution TL but takes just a fraction of the runtime. Our progress has been substantial, and our improvements will enable the generated TLs to be of sufficient quality to be useful to warfighters in a wide range of cases. More importantly, these conclusions are applicable to all of our physics domains.

Our work focused on utilizing approximate solutions with significantly reduced computational requirements that can be computed dynamically to improve generated transmission losses (TLs) in cases where the original PALIS code produced results of insufficient quality. Experiments demonstrated this method's effectiveness, leading to PALIS code modifications to accept multi-channel inputs. We determined that acoustic ray-traces as secondary input channels perform equivalently to or better than low-resolution TL inputs while requiring substantially reduced runtime. This progress enables generating TLs of sufficient quality for operational warfighter applications across diverse scenarios.

Our additional experiments investigated using secondary networks to input ray-trace vectors instead of secondary input channels, providing a simpler and faster implementation. Results demonstrated that ray-trace vectors performed equivalently to secondary channels, becoming our default PALIS operation method.

2.2 Cyclical generation:

Physics emulator projects aim to produce accurate simulation results from environmental inputs with significant speed advantages over physics simulators. While PALIS successfully achieves this goal in most cases, we investigated a critical question: **determining generated result trustworthiness (TLs in PALIS) without comparison to physics simulator outputs.**

We explored an approach involving complete generation cycles: generating TLs from SSPs and subsequently generating SSPs from TLs. By comparing generated SSPs to actual input SSPs, we hypothesized we could assess generated TL accuracy – if the generated SSPs closely resembled original SSPs, we could accept generated TLs as accurate. Cyclical generation code was completed in FY24, with experimentation conducted in FY25.

Our results indicated this method could not effectively evaluate generated TL quality. Even poorly generated TLs produced high-quality SSPs. Analysis revealed this occurred due to TL complexity versus SSP simplicity, making neural network learning from TLs to SSPs substantially easier than SSP-to-TL learning. Confirmation experiments using smaller architectures for SSP generation from TLs demonstrated that simple models with poor-quality TLs could generate high-quality SSPs.

Our current efforts focus on training separate, non-transformer models to generate ray-traces from TLs, since ray-trace vector complexity matches TL complexity while maintaining rapid computation. These experiments are ongoing.

2.3 Data Preparation

PALIS accepts 256×256 pixel images with values ranging from 0 to 1 as inputs and generates corresponding output images. Input images represent Sound Speed Profiles (SSP), while outputs represent Transmission Loss (TL) values at each pixel location, derived from Navy Standard Parabolic Equation (NSPE) calculations. Since NSPE output TL values exist on depth and range grids that do not necessarily correspond to 256×256 grids, TL values undergo interpolation to fit this format.

The data organization reflects TL and SSP dependence on local environmental parameters, including ocean state and bathymetry. Datasets represent different ocean regions, with each dataset divided into training and test sets (approximately 90% and 10%, respectively). TL values also depend on source depth and sound frequency. For each SSP, TL images were generated across various source depths and frequencies. Most datasets included source depths of 10m and 100m, with frequencies of 50, 100, 200, 500, and 1000 Hz. Special datasets with randomized parameters were created to enable model generalization to arbitrary source depths and frequencies.

Fixed range and depth grids present challenges in shallow water cases. Below bathymetry depths, TL values become large and map to 1, resulting in images predominantly composed of 1s with limited dynamic areas near surfaces. Shallow water cases typically contain more acoustic features, such as bottom bounces. To address this issue, images undergo "scaling" to new grids with the deepest geological SSP (approximately 500m below bathymetry) as the final row. While the final column typically remains unchanged, upslope conditions require selecting the last range value before bathymetry reaches 0m as the final column. Resulting grids smaller than 256×256 undergo interpolation back to full resolution. Although this process introduces some image degradation, the benefits of increased dynamic pixel content outweigh the interpolation disadvantages.

2.4 Performance measurements/metrics:

Following PALIS model training on datasets, evaluation occurs on separate test datasets to assess performance. Each test TL output undergoes comparison with the corresponding original TLs using multiple metrics: Mean Squared Error (MSE), Structural Similarity Index (SSIM), and Fréchet Inception Distance (FID). These metrics generally provide consistent rankings of input/output comparisons, with SSIM serving as the preferred metric due to its balance between computational efficiency and descriptiveness. Following SSIM-based ranking, samples of worst, best, and intermediate cases undergo visualization using Python, with original and model outputs displayed comparatively. Acoustics experts review these visualizations to verify model improvements and identify potential physics content issues.

Ray-trace and bathymetry incorporation into SSP input images improved model capability for capturing fine TL output details. Prior to this addition, PALIS occasionally added or omitted essential acoustic features such as bottom bounces. Ray-trace feature provision enables more accurate feature reproduction. Future work will incorporate additional physics-based information to further enhance results.

2.5 Generative Adversarial Network (GAN) architecture

Given the diverse strengths and limitations of various network architectures for generative applications, exploring and comparing multiple options remains essential. Network performance significantly depends on dataset characteristics, motivating continued experimentation with novel architectures in FY25.

The primary motivation involved training generators using discriminator loss functions. Our hypothesis suggested that this training would enable models to distinguish between generated and real TLs at sound source frequencies not utilized in pretraining. The GAN approach should allow models to learn loss field generation across all source frequencies, analogous to GANs generating novel images not included in training datasets.

During FY25, we created the necessary PALIS data for GAN training and implemented/tested the GAN architecture. However, experiments across all frequencies remain ongoing and were interrupted to focus on neural operator approaches described in Section 3.1, as these represent more promising methodological directions.

3. Meteorology

In FY24, our meteorology point of contact departed NRL, and the Meteorology Division superintendent decided to defund our project for FY25 and FY26. Although this decision prevented additional progress, FY24 results were substantial, leading to the initial FY25 months being dedicated to manuscript preparation for open literature publication.

Our meteorology domain work advanced Navy electromagnetic warfare capabilities by developing atmospheric propagation prediction systems. Experiments yielded preliminary data supporting our system's rapid predictive accuracy for radar pattern propagation. We prepared, revised, and submitted the manuscript "Pattern Propagation Factor Prediction Using A Deep Neural Network Generator" to IEEE Antennas and Wireless Propagation Letters, a leading scientific journal. This paper remains under peer review. All work supported Navy electromagnetic spectrum superiority objectives.

4. Structural health monitoring

This task domain aims to generalize real-time neural network systems for monitoring material damage on naval platforms by utilizing laboratory methods from the 6.1 MINC program. Baseline experiments were performed at room temperature, with generalization goals to account for large temperature fluctuations (-60°C to 40°C). Beyond interpolation between temperature extremes, we will analyze data imbalance issues and neural network generalization capabilities. COMSOL simulations are configured for temperature extremes and random temperature distributions between them. COMSOL licensing and server cluster deployment issues delayed progress. These issues have been resolved, and data generation awaits server availability.

References

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Keywords

Neural network generalization, Neural operators, Wave propagation modeling, Helmholtz equation, Scientific machine learning, Deep learning for physics/PDE solvers, Underwater acoustics, Electromagnetic propagation, Structural health monitoring (SHM)

Dual Use

This research is expected to yield substantial reductions in computational resource requirements, with projected acceleration factors of two to three orders of magnitude in physics simulations compared to traditional numerical methods. Additionally, enhanced generalization capabilities across diverse physical environments should substantially improve system accuracy in complex and varied operational scenarios. Enhanced sonar prediction accuracy has the potential to improve ASW effectiveness, while real-time EM propagation models may provide capabilities to counter adversary electronic warfare (EW) tactics. While focused on Navy systems, this work has potential applications benefiting other DOD services and civilian maritime industries.

Plans

FY26

We will finish the NSNO code implementation, and the next step will be to conduct generalization experiments on the operator network. Given the success of this method, we plan to extend the model to more complex physics found in real oceans, such as frequency-dependent attenuation, currents, and non-linear effects. We plan to implement hybrid methods, such as incorporating analytical solutions of Green's function to handle source singularities and using the NOs to solve for the residual function.

We plan to build on our work with acoustics to create a neural operator implementation for the SHM application, where a wave traveling through metal will indicate if there is a defect in the material. Given the success of this method, we plan experiments to test real-world complexities, such as hybrids of inhomogeneous materials, varying temperatures, and external noise.

We plan to continue our work in measuring the uncertainty of the generated results and provide a way for the user to know whether and when to trust the results.